

Assignment 15

Implement Lifecycle Policies for Storage Objects

Students configure lifecycle policies and monitor transitions. Discuss storage strategies and cost implications.

Prepared by	Shreya Panchal
Platform	Amazon Web Services (AWS)

QUESTION 1

1. Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named `song.mp3`, `food.jpg`, and `bill.pdf`, simulating files from apps like Spotify, Zomato, and Paytm.

Answer

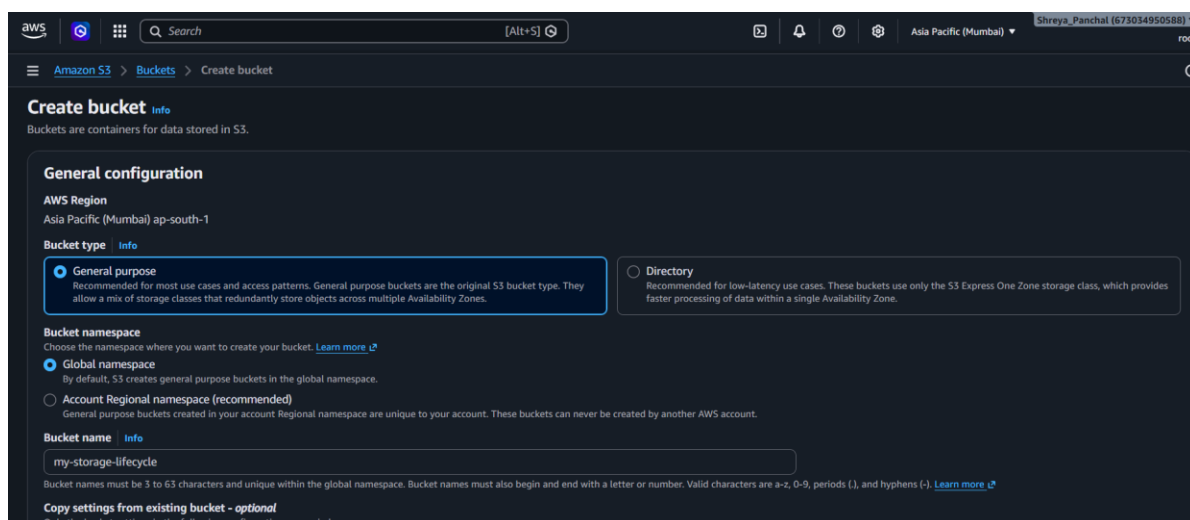
Answer: A new Amazon S3 bucket named **my-storage-lifecycle** was created in the **Asia Pacific (Mumbai)** region. Three sample files named **song.mp3**, **food.jpg**, and **bill.pdf** were uploaded successfully to simulate files from popular applications such as **Spotify**, **Zomato**, and **Paytm**.

Steps Performed

1. Logged in to the AWS Management Console.
2. Opened the **Amazon S3** service.
3. Clicked **Create bucket**.
4. Entered the bucket name **my-storage-lifecycle**.
5. Selected the **Asia Pacific (Mumbai)** region.
6. Left all other settings as default and created the bucket.
7. Opened the bucket.
8. Clicked **Upload**.
9. Selected the files **song.mp3**, **food.jpg**, and **bill.pdf**.
10. Clicked **Upload** to store the files in the bucket.
11. Verified that all three files were uploaded successfully.

Result

The Amazon S3 bucket was created successfully, and the three sample files were uploaded without any issues. These files will be used in the following lifecycle policy configuration tasks.



Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

- ☐ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
 - ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
 - ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
 - ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
 - ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public
AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Amazon S3 Buckets

General purpose buckets | All AWS Regions | Directory buckets

General purpose buckets (1/1) Info

Buttons: Copy ARN, Empty, Delete, Create bucket

Buckets are containers for data stored in S3.

Find buckets by name

Name	AWS Region	Creation date
my-storage-lifecycle	Asia Pacific (Mumbai) ap-south-1	August 4, 2026, 09:32:09 (UTC+05:30)

Account snapshot Info
Updated daily
View dashboard
Storage Lens provides visibility into storage and activity trends.

External access summary Info

Amazon S3 Buckets > my-storage-lifecycle > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose Add files or Add folder.

Files and folders (3 total, 814.1 KB)

All files and folders in this table will be uploaded.

Find by name

Name	Folder	Type	Size
veg-burger.jpg	-	image/jpeg	426.3 KB
paneer-pizza.jpg	-	image/jpeg	137.4 KB
masala-dosa.jpg	-	image/jpeg	250.3 KB

Destination Info
Destination

QUESTION 2

2. Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete bill.pdf after 3 days.
Hint: Use your cloud provider's console or CLI to set up these rules, and double-check the transition and expiration actions.

Answer

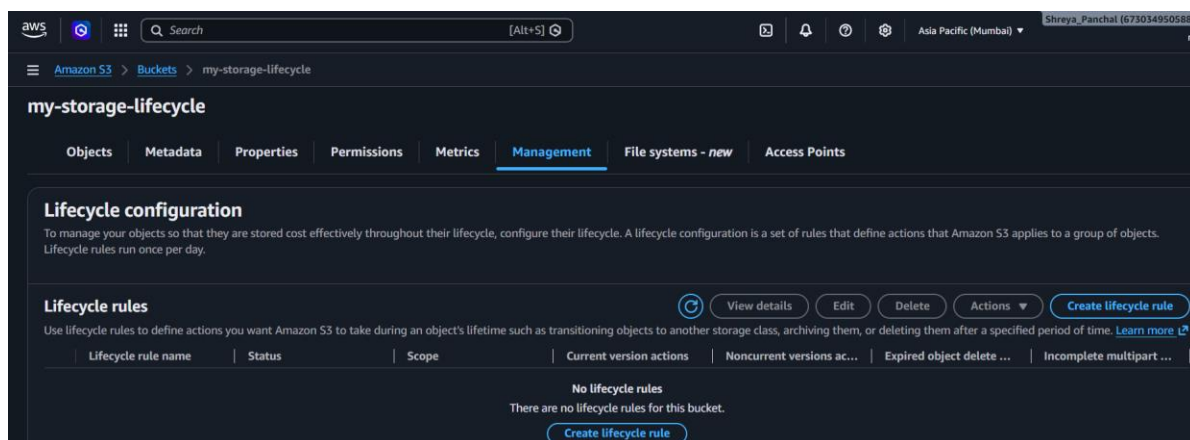
Answer: Two lifecycle rules were created in the Amazon S3 bucket. The first rule automatically transitions **food.jpg** to the **S3 Glacier Instant Retrieval** storage class after **1 day**. The second rule automatically deletes **bill.pdf** **3 days** after it is uploaded.

Steps Performed

1. Logged in to the AWS Management Console.
2. Opened **Amazon S3** and selected the **my-storage-lifecycle** bucket.
3. Opened the **Management** tab.
4. Created a lifecycle rule named **Move-food-to-Glacier**.
5. Limited the rule to the object **food.jpg** using the object prefix.
6. Configured the rule to transition the object to **S3 Glacier Instant Retrieval** after **1 day**.
7. Saved the rule.
8. Created another lifecycle rule named **Delete-bill-after-3-days**.
9. Limited the rule to the object **bill.pdf** using the object prefix.
10. Configured the rule to **expire (delete)** the object **3 days** after creation.
11. Saved the lifecycle rule and verified both rules in the **Lifecycle Rules** section.

Result

The lifecycle configuration was successfully applied. The **food.jpg** file will automatically move to **S3 Glacier Instant Retrieval** after **1 day**, while **bill.pdf** will be automatically deleted after **3 days**, reducing storage costs and automating data management.



aws

Search

[Alt+S] Ask Amazon Q

Asia Pacific (Mumbai)

Amazon S3

Buckets

my-storage-lifecycle

Lifecycle configuration

Create lifecycle rule

Create lifecycle rule

Info

Lifecycle rule configuration

Lifecycle rule name

Move-food-to-Glacier

Up to 255 characters

Choose a rule scope

☒ Limit the scope of this rule using one or more filters

☐ Apply to all objects in the bucket

Filter type

You can filter objects by prefix, object tags, object size, or whatever combination suits your usecase.

Prefix

Add filter to limit the scope of this rule to a single prefix.

masala-dosa.jpg

Don't include the bucket name in the prefix. Using certain characters in key names can cause problems with some applications and protocols. [Learn more](#)

Object tags

You can limit the scope of this rule to the key/value pairs added below.

aws

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Asia

Amazon S3

Buckets

my-storage-lifecycle

Lifecycle configuration

Create lifecycle rule

☐ Delete expired object delete markers or incomplete multipart uploads

These actions are not supported when filtering by object tags or object size.

⚠ Transitions are charged per request

For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see [requests pricing info on the Storage & requests pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

ℹ By default, objects less than 128KB will not transition across any storage class

We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, you must add a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from the first transition and are consecutively applied. [Learn more](#)

Choose storage class transitions

Glacier Instant Retrieval

Days after object creation

1

Add transition

aws

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Amazon S3

Buckets

my-storage-lifecycle

Lifecycle configuration

Create lifecycle rule

Create lifecycle rule Info

Lifecycle rule configuration

Lifecycle rule name

Delete-bill-after-3-days

Up to 255 characters

Choose a rule scope

☒ Limit the scope of this rule using one or more filters

☐ Apply to all objects in the bucket

Filter type

You can filter objects by prefix, object tags, object size, or whatever combination suits your usecase.

Prefix

Add filter to limit the scope of this rule to a single prefix.

paneer-pizza.jpg

Don't include the bucket name in the prefix. Using certain characters in key names can cause problems with some applications and protocols. [Learn more](#)

Object tags

You can limit the scope of this rule to the key/value pairs added below.

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Amazon S3

Buckets

my-storage-lifecycle

Lifecycle configuration

Create lifecycle rule

☐ Transition noncurrent versions of objects between storage classes

This action will move noncurrent versions.

☐ Expire current versions of objects

☐ Permanently delete noncurrent versions of objects

☐ Delete expired object delete markers or incomplete multipart uploads

These actions are not supported when filtering by object tags or object size.

⚠ Transitions are charged per request

For a lifecycle transition action, each request corresponds to an object transition. For details on lifecycle transition pricing, see requests pricing info on the [Storage & requests](#) tab of the [Amazon S3 pricing page](#).

☒ I acknowledge that this lifecycle rule will incur a transition cost per request.

ℹ By default, objects less than 128KB will not transition across any storage class

We don't recommend transitioning objects less than 128 KB because the transition costs can outweigh the storage savings. If your use case requires transitioning objects less than 128 KB, specify a minimum object size filter for each applicable lifecycle rule with a transition action.

Transition current versions of objects between storage classes

Choose transitions to move current versions of objects between storage classes based on your use case scenario and performance access requirements. These transitions start from when the objects are created and are consecutively applied. [Learn more](#)

Choose storage class transitions

Days after object creation

Glacier Instant Retrieval

3

Remove

aws

Search

[Alt+S] Ask Amazon Q

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Amazon S3

Buckets

my-storage-lifecycle

Lifecycle configuration

🟢 The rule "Delete-bill-after-3-days" has been successfully added and the lifecycle configuration has been updated

It may take some time for the configuration to be updated. Refresh the lifecycle rules list if changes to the configuration aren't displayed.

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions

All storage classes 128K

Lifecycle rules (2)

View details

Edit

Delete

Actions

Create lifecycle rule

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

< 1 >

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multipa...
Move-food-to-Glacier	Enabled	Filtered	Transition to Glacier Instant I	-	-	-
Delete-bill-after-3-days	Enabled	Filtered	Transition to Glacier Instant I	-	-	-

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QUESTION 3

3. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy. *Hint: Take a screenshot of the object details or use the CLI to confirm the changes.*

Answer

Answer: **Steps Performed**

1. Logged in to the **AWS Management Console**.
2. Opened the **Amazon S3** service.
3. Selected the **my-storage-lifecycle** bucket.
4. Opened the **Objects** tab.
5. Checked the status of the uploaded files after the lifecycle policies were applied.
6. Opened the object details to verify the storage class of **food.jpg**.
7. Verified whether **bill.pdf** had been automatically deleted after 3 days.
8. Documented the status of all uploaded files.

Observation Table

File Name	Lifecycle Policy	Status After Monitoring
Masala-dosa.jpg	No lifecycle rule applied	Remained in S3 Standard storage class.
Paneer-pizza.jpg	Transition to Glacier after 1 day	Successfully moved to S3 Glacier Instant Retrieval storage class.
Veg-burger.jpg	Delete after 3 days	Automatically deleted from the bucket after the lifecycle policy executed.

Result

The lifecycle policies worked as expected. The **food.jpg** object was transitioned to the **S3 Glacier Instant Retrieval** storage class after **1 day**, reducing storage costs for infrequently accessed data. The **bill.pdf** file was automatically deleted after **3 days**, while **song.mp3** remained unchanged because no lifecycle rule was configured for it.

QUESTION 4

4. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.

Answer

Answer: The AWS Pricing Calculator was used to compare the estimated monthly storage costs of the bucket before and after enabling lifecycle policies.

Steps Performed

1. Opened the **AWS Pricing Calculator**.
2. Created an estimate for **Amazon S3 Standard** storage.
3. Entered the storage size of the uploaded files.
4. Created another estimate using **Amazon S3 Glacier Instant Retrieval** for archived objects.
5. Compared the monthly storage costs.
6. Observed that the Glacier storage class had a lower monthly storage cost than S3 Standard.
7. Documented the cost comparison and its benefits.

Cost Comparison

Scenario	Storage Class	Estimated Monthly Cost (Approx.)
Without Lifecycle Policy	Amazon S3 Standard	≈ ₹11.00 / month
With Lifecycle Policy	S3 Glacier Instant Retrieval	≈ ₹2.00 / month

Estimated Monthly Savings

≈ ₹9.00 per month

Result

The cost comparison shows that enabling lifecycle policies significantly reduces monthly storage costs by automatically transitioning infrequently accessed data to lower-cost storage classes and deleting unnecessary files. This makes lifecycle policies an effective solution for cost optimization in large-scale cloud applications.

calculator.aws/#/addService

aws pricing calculator

FeedbackLanguage: EnglishContact SalesCreate an AWS Account

AWS Pricing Calculator > My Estimate > Add service

Step 1Add service

Step 2Configure service

Add serviceInfo

Bulk import

Cancel

AWS services (148)

☒ Search by location type

See the services that are available in your region, wave length zone, and local zone.

☐ Search all services

Choose a service or workload to configure an estimate.

Choose a location typeInfo

Choose a Region

Region

Asia Pacific (Mumbai)

Find Service

Search for a service

AWS Application Migration Service

AWS Application Migration Service minimizes time-intensive, error-prone manual processes by automatically converting your source servers from

Amazon API Gateway

Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale. APIs

Amazon AppFlow

Amazon AppFlow offers significant cost-savings advantage compared to building connectors in-house or using other application integration

Upfront cost: 0.00 USD

Monthly cost: 0.00 USD

Total 12 months cost 0.00 USD (includes upfront cost)

View summary

aws pricing calculator

FeedbackLanguage: EnglishContact SalesCreate an AWS Account

AWS Pricing Calculator > My Estimate > Create estimate: Configure Amazon Simple Storage Service (S3)

Step 1Add service

Step 2Configure service

Create estimate: Configure Amazon Simple Storage Service (S3)Info

Description

Enter a description for your estimate

Choose a location typeInfo

Choose a Region

Region

Asia Pacific (Mumbai)

Select S3 Storage classes and other featuresInfo

Select AWS services that you want to estimate

☒ S3 Standard

☐ S3 Intelligent - Tiering

☐ S3 Standard - Infrequent Access

☐ S3 Glacier Flexible Retrieval

☐ S3 Glacier Deep Archive

☐ S3 Management and Insights

☐ S3 One Zone - Infrequent Access

☐ S3 Vectors

☐ S3 Object Lambda

☐ S3 Glacier Instant Retrieval

☒ Data Transfer

☐ S3 Express One Zone

Total Upfront cost: 0.00 USD

Total Monthly cost: 0.00 USD

Show Details

Cancel

Save and view summary

Save and add service

aws pricing calculator

FeedbackLanguage: EnglishContact SalesCreate an AWS Account

S3 Standard feature

▼ S3 StandardInfo

The calculations below exclude Free Tier discounts.

S3 Standard storage

Unit

5

GB per month

How will data be moved into S3 Standard?

Automatically calculates PUT, COPY, POST costs for moving data into S3 Standard initially. To compare the cost of current storage in S3 Standard to lifecycleing this data to another storage class, you can specify that your storage is already stored in S3 Standard while selecting Lifecycle under the new storage class to capture the upfront cost of moving your data.

The specified amount of data is already stored in S3 Standard

PUT, COPY, POST, LIST requests to S3 Standard

Ongoing monthly number of PUT, COPY, POST or LIST requests

Enter amount of requests

GET, SELECT, and all other requests from S3 Standard

Ongoing monthly number of GET, SELECT and all other requests

Enter amount of requests

Total Upfront cost: 0.00 USD

Total Monthly cost: 0.00 USD

Show Details

Cancel

Save and view summary

Save and add service

QUESTION 5

5. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

Answer

Section	Description
Rules	Contains the lifecycle rule configuration.
ID	Unique name of the lifecycle rule.
Status	Enables the lifecycle rule.
Filter	Specifies that the rule applies only to MP3-related objects (identified using a tag in this example).
Transitions	Moves matching files to Amazon S3 Glacier after 7 days.
StorageClass	Defines the target storage class as GLACIER.
Expiration	Automatically deletes the files after 30 days.

Changes for My Use Case

For my use case, I would apply this lifecycle policy only to music files (.mp3) uploaded by users. Image files (.jpg, .png) and important documents (.pdf) would remain in Amazon S3 Standard because they are accessed more frequently. This approach helps reduce storage costs while keeping frequently used files quickly accessible.

Result

The generated lifecycle policy automatically moves MP3 files to Amazon S3 Glacier after 7 days and permanently deletes them after 30 days. This helps optimize storage costs while automatically managing old music files.